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Machine Learning Assisted History Matching Application in the North Kuwait Jurassic

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Abstract

History Matching (HM) is a critical process in reservoir simulation and management, aiming to calibrate reservoir models to historical production data to predict future performance and guide development decisions. However, traditional HM processes are often time-consuming due to the complexity of reservoir systems and rely on manual trial-and-error approaches. In North Kuwait's Jurassic Fields, an annual manual HM process across 15 reservoir models has been applied historically which has been time-intensive with limited range of uncertainties.

The increasing complexity of reservoirs, characterized by a large number of parameters and complicated parameter-response relationships, necessitates innovative approaches. To address these challenges, this work introduces an integrated framework, a novel approach, combining Assisted History Matching (AHM) and Machine Learning (ML) techniques to achieve efficient and reliable history matching, enhance probabilistic forecasting, and improve well-placement strategies.

By leveraging the power of ML, we address the challenges associated with large-scale, complex reservoir models and high uncertainty. Our proposed methodology automates the parameter calibration process, reducing the reliance on human intervention and significantly accelerating the HM workflow. This approach ensures the preservation of geological complexity while considering uncertainties in subsurface parameters. By coupling AHM with ML, we not only expedite the history-matching process, achieving results up to 5–10 times faster depending on model size and complexity, but also deliver ensemble-based history-matching outputs that enable probabilistic forecasts. This facilitates the identification of infill drilling locations based on robust probabilistic assessments, improving decision-making under uncertainty.

Two case studies from the North Kuwait Jurassic Field demonstrates the effectiveness of our approach. Machine learning-assisted history matching was applied to a Jurassic carbonate reservoir with complex heterogeneities and high subsurface uncertainties. By employing advanced ML algorithms, the workflow efficiently calibrated the reservoir simulation model to historical production data, providing a better match to observed field performance compared to conventional methods. The resulting probabilistic forecast models captured the full spectrum of uncertainties, enabling more reliable predictions of future reservoir behavior [4].

Furthermore, the outputs of ML-assisted history matching can seamlessly be integrated into an automated machine learning framework [3] for well-placement optimization. This automated process identifies optimal locations for infill wells, considering both geological and operational constraints. The integration of probabilistic forecasts ensures that well-placement decisions are aligned with the full range of possible reservoir scenarios, reducing the risk of suboptimal field development.

Introduction

Reservoir models are essential tools for predicting hydrocarbon production and optimizing field operations [1]. However, their accuracy is often compromised by uncertainties in reservoir parameters. Traditional history matching (HM) techniques have been employed to improve model calibration [2], but they often rely on manual parameter adjustments, which can be time-consuming and fail to capture complex nonlinear relationships between variables. These limitations motivated the exploration of machine learning (ML) approaches to improve HM workflows.

In recent years, machine learning (ML) has emerged as a promising approach for improving HM workflows. ML algorithms can efficiently optimize complex functions, reducing the need for manual parameter tuning and enabling real-time data assimilation. However, a critical challenge in integrating ML with reservoir simulators lies in selecting appropriate hyperparameter optimization techniques that balance performance and computational efficiency. In this study, we address this challenge by applying machine learning-hybrid history matching (ML-AHM) to the North Kuwait Jurassic reservoir, a field known for its complex geology and uncertain reservoir properties and demonstrated its potential for improving model calibration and predictions. Our methodology involved integrating a machine learning algorithm with an existing reservoir simulation model, enabling direct data assimilation and optimization of reservoir parameters. We conducted extensive hyperparameter tuning using Bayesian optimization, ensuring optimal performance of the ML-HM workflow. This study aims to demonstrate the application of ML-HM to the North Kuwait Jurassic Middle Marrat reservoir, a challenging field characterized by complex geology and uncertain reservoir properties.

Our results indicate significant improvements in model calibration, with a reduction of up to 50% in the objective function and a corresponding decrease in root mean square error (RMSE) for both pressure and production fits. Furthermore, our analysis revealed that the integrated AI-driven workflow can effectively capture nonlinear relationships between reservoir parameters, leading to more accurate predictions.

The significance of this study lies in its demonstration of the applicability of ML-AHM to real-world petroleum engineering problems. Our approach not only improves model accuracy but also enhances computational efficiency and reduces uncertainty associated with traditional history matching methods.

Field Background

The North Kuwait Jurassic (NKJ) fields are a complex series of deep onshore carbonate plays with intensely heterogeneous reservoir properties. The main reservoir unit is the Middle Jurassic Marrat Formation (limestone and dolomite), which exhibits low primary porosity, low matrix permeability, and high fracture density. This heterogeneity is further complicated by the dual-porosity nature of these fields, where secondary porosity (vugs, fractures) varies both laterally and vertically. Two NKJ fields, Field A and Field D, were used for this study work:

Field (D) North Kuwait Jurassic Field Background

In the Field D, an isolated anticlinal closure in the Marrat carbonate sequence is observed. The reservoir section is predominantly Middle Marrat limestone/dolomite. Low primary porosity and extremely low matrix permeability are evident, with production sourced from natural fractures and local dolomitization.

Field (A) North Kuwait Jurassic Field Background

Field A area lies further north than Field D. It is expected to have similar reservoir characteristics (porous dolomite zones and vugs, fracture-dominated deliverability). Field A is characterized by high initial reservoir pressures (~10-11 kpsi) and elevated temperatures.

Reservoir Property Uncertainty in North Kuwait Jurassic Fields

The North Kuwait Jurassic fields exhibit pronounced anisotropy, with matrix permeability often falling below 0.1 mD in tighter zones [2]. This introduces large uncertainties in simulated well rates and pressures, making accurate reservoir modeling a significant challenge. Furthermore, HPHT conditions in tight zones and multi-phase (oil/gas/water) flow in main producing zones, with pressure depletion below saturation pressure, further complicate PVT and relative-permeability estimations, inflating the range of plausible reservoir-property realizations.

- Porosity: Low primary porosity (~a few percent) with secondary porosity from vugs and dolomitization.
- Permeability: Low matrix permeability with variable fracture density.
- Fluids: Varying from black high volatile oil to condensate and sour gas, with production coming from both fine limestone matrix and secondary fractures/dolomite porosity.

Challenges in Reservoir Simulation and History Matching

Conventional gradient-based or trial-and-error history match relies on local sensitivities and manual adjustment of rock and fluid properties. The assisted history match is introduced to improve the process of history match by the definition of an objective function comprising pressure and rates misfits become a common practice; however, it frequently fails to converge when faced with non-Gaussian parameter distributions and numerous local minima. This challenge becomes even more pronounced in scenarios where multiple wells need to be matched simultaneously. In such cases, the process can become intractable, resulting in model ensembles that continue to diverge significantly in their results accuracy and performance forecasts.

The inability to achieve consistent results has significant implications for reservoir management. Why does this matter? The accurate prediction of fluid flow and pressure behavior is crucial for informed decision-making and optimal resource recovery. In the absence of reliable models, engineers are forced to rely on manual tuning, which can lead to inconsistent outcomes and reduced confidence in simulation results.

Why ML-Assisted History Matching is Needed

The existing challenges in D and A field reservoir simulation models and history matching highlight the need for more accurate and reliable methods to reduce uncertainty in reservoir properties. The complexity of the permeability field, with anisotropic and patchy behavior, requires careful calibration of fracture and matrix properties.

- High uncertainty in permeability field due to anisotropic and patchy behavior.
- Dual-porosity (matrix plus fracture) behavior requires careful calibration of fracture and matrix properties.

Advanced tools, such as assisted history match and machine learning, are necessary for history matching.

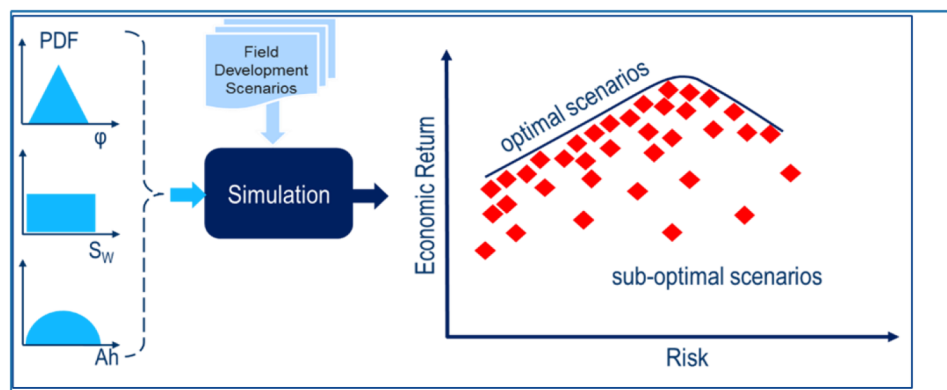


Figure 1—Subsurface uncertainties influence the economical return of field developments scenarios. (better to remove this map)

By reducing uncertainty and improving the accuracy of reservoir models, ML-assisted history matching can help optimize production strategies, reduce risks, and improve overall field performance. The ability to identify patterns and relationships between different variables through machine learning algorithms can lead to more efficient and effective calibration of reservoir properties:

- **Enhanced Uncertainty Quantification:** ML-derived posterior distributions capture higher-order statistics of reservoir properties, tightening predictive intervals for well performance.
- **Improved Computational Efficiency:** Surrogate models help to cut overall simulation runs by up to 50%, enabling sensitivity studies and iterative design of well placements.

Methodology

General Workflow of Machine Learning- Assisted History Match is shown in figure 2. It consists of two major parts. The first part consists of typical uncertainty and optimization workflow and the second part consists of integration of machine learning for optimum parameters identification.

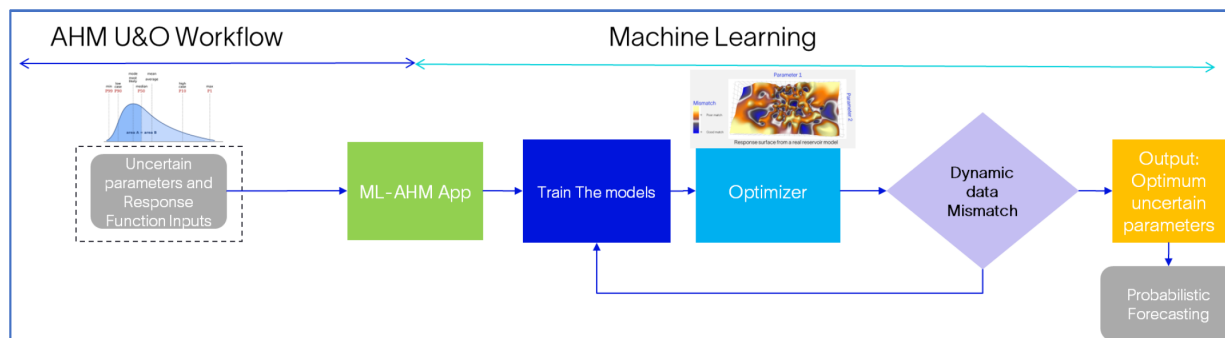


Figure 2—General Workflow of Machine Learning- Assisted History Match

Assisted History Matching

Assisted History Matching (AHM) is an automated reservoir-model calibration process that uses optimization algorithms to adjust uncertain parameters and reconcile simulated reservoir outputs with observed production and pressure data [2]. By defining an objective function based on differences between measured and modeled rates, pressures, water-cuts, etc., The misfit can be minimized by varying properties such as permeability multipliers, porosity, relative-permeability, fault transmissibility, or aquifer size. It combines human expertise with computational tools to adjust model parameters systematically, improving both the speed and quality of the history matching. Key benefits of AHM include automated reservoir-model

calibration process, systematic framework for exploring a broad parameter space, integration of multiple data types, and generation of multiple history-matched models (realizations) to quantify uncertainty. In practice, AHM has several advantages over manual tuning:

- Rapid convergence on history-matched models
- Quantification of parameter uncertainty
- Reduced project timelines
- Identification of key geological factors impacting performance

However, assisted history matching algorithms have some limitations. For instance:

- They may fall into local minimum, failing to identify the global optimum.
- The complex parameter space can be challenging to navigate.
- Simple proxy relationships cannot capture the complexity of real reservoirs.

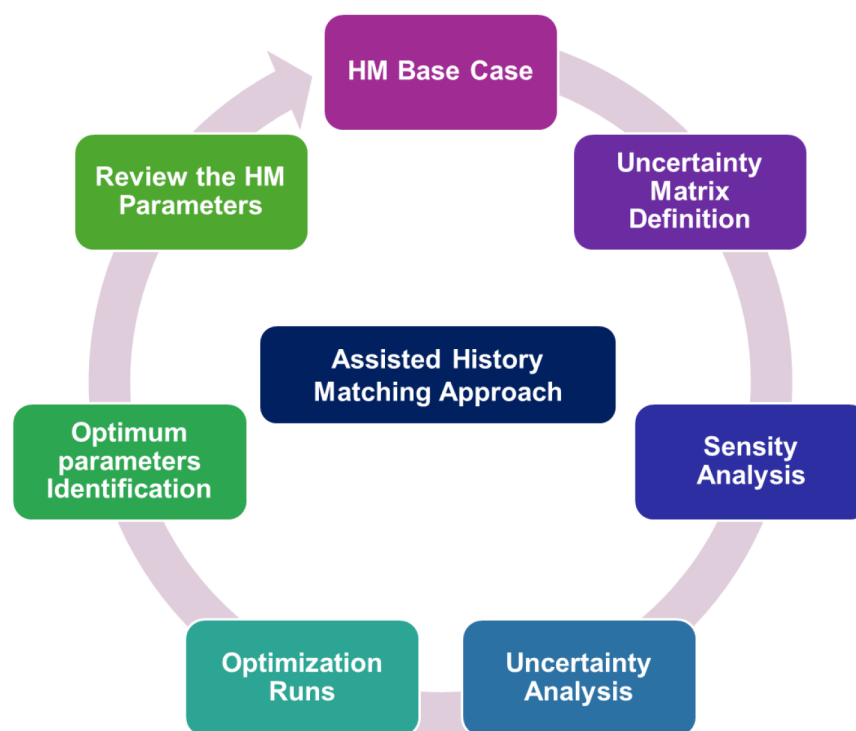


Figure 3—Assisted History Workflow prior to ML integration

To address these challenges, integrating machine learning with assisted history match is beneficial. Machine learning techniques can help overcome limitations such as:

- Identifying multiple parameter combinations that improve history matching
- Capturing highly non-linear response surfaces
- Avoiding local minimum and identifying the global optimum

Machine Learning Assisted History Match

ML-AHM methodology leverages the outputs of multiple realization reservoir uncertainty simulations, generated by varying subsurface uncertainty parameters, to train machine learning proxy models to reproduce the response of the reservoir simulation model [1]. These proxy models then serve as a surrogate for the physics-based simulator, facilitating computer-assisted history matching. A dedicated optimization framework is applied to identify the optimal set of uncertainty parameters that minimize the discrepancy between simulated and actual production and pressure data. The resulting parameters are subsequently utilized to update the reservoir simulation model that is run through the physics-based reservoir simulator.

This approach has the potential to significantly improve the accuracy and reliability of subsurface models by combining the strengths of both approaches AHM and ML. In addition, the benefits of integrating machine learning with assisted history match include:

- **Efficiency:** Reduces the time required to calibrate models compared to manual history matching.
- **Scalability:** Handles complex models with many uncertain parameters.
- **Improved Accuracy:** Optimization algorithms can identify parameter combinations that might be overlooked in manual matching.

By harnessing the capabilities of machine learning algorithms, it is possible to efficiently explore a vast parameter space and identify optimal solutions that may not be feasible with traditional methods. However, it is essential to exercise caution when relying on proxy models, as they can introduce biases and errors if not properly trained and calibrated

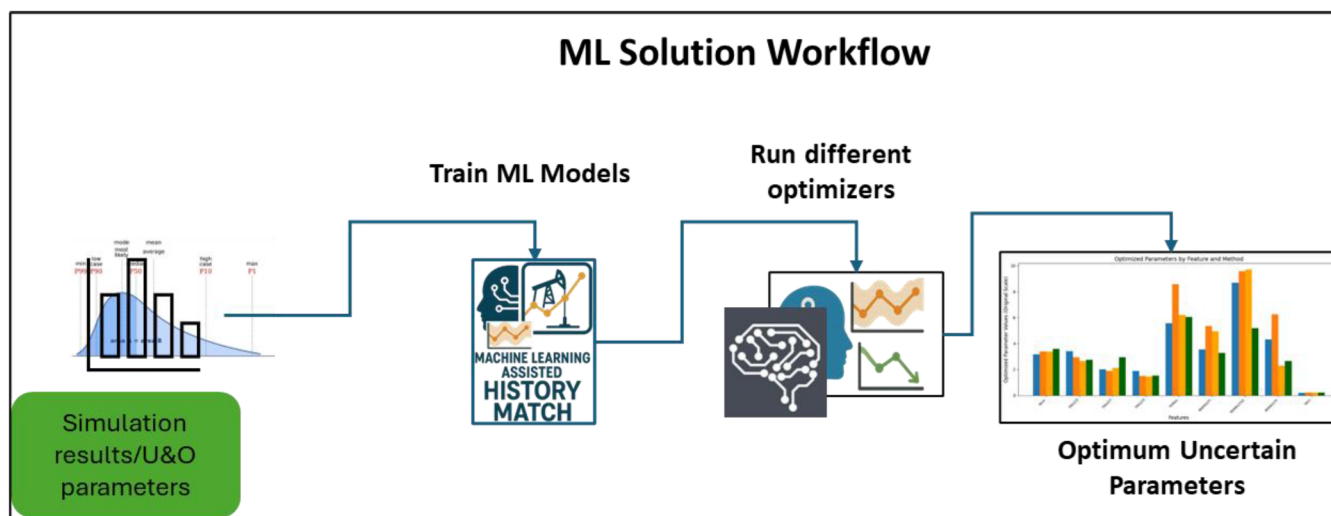


Figure 4—ML-AHM High-level Architecture

Case Study

This case study demonstrates the application of machine learning (ML) techniques to improve history matching in two challenging fields D and A of North Kuwait Jurassic Complex. Both fields exhibit unique geological features, such as low primary porosity, low matrix permeability, and fracture-dominated deliverability (D field). Despite these complexities, accurate modeling of reservoir behavior is crucial for optimizing field development plans, predicting production rates, and identifying opportunities for improved recovery. However, traditional history matching approaches often struggle to capture the complexities of these fields. In this case study, we explore the application of ML-assisted history matching to improve model accuracy and robustness in D and A Fields. By leveraging advanced ML algorithms and techniques, we

aim to enhance our understanding of these complex reservoirs and develop more reliable predictions for field performance.

Data Sources and Preliminary Analysis

We utilized production (gas, oil, water) rates, tubing head pressure, reservoir pressure, and bottom-hole pressure (BHP) data from production wells, alongside reservoir properties. Static geomodels provided grid-block properties: Static parameters included porosity, permeability anisotropy, and relative permeability endpoints. Dynamic simulation outputs were matched to historical data points at daily intervals.

Definition of uncertainty matrix. The first step of the assisted history match workflow is to define and determine which data is indeterminate and the outcomes are captured in the uncertainty matrix, [figure 5](#). It was used to quantify and manage both static and dynamic uncertainties of D and A fields. The uncertainty matrix is based on an integrated team effort and includes input from various knowledge domains.

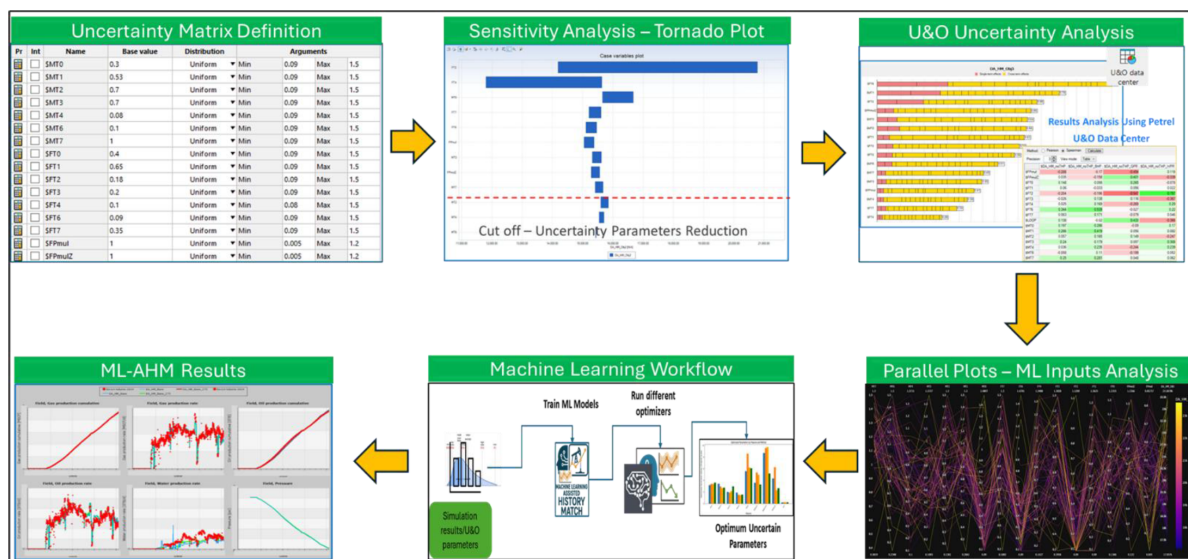


Figure 5—ML - Assisted history matching process

Variables considered for assisted history match study are matrix transmissibility, fracture transmissibility, horizontal Permeability, vertical permeability, skin factor, relative permeability end points, and well inflow parameters which are considered to be indeterminate (high uncertainty). The data to be matched and its tolerances are captured in an objective function.

An accurate uncertainty matrix ensures that history matches are robust and reliable, reducing the likelihood of overfitting or underfitting. Furthermore, by propagating uncertainties through models, it helps to better communicate risks and opportunities to stakeholders. This enhanced transparency fosters more informed decision-making throughout the asset life cycle.

Sensitivity analysis. To know which variables may have a significant effect on the history match and which may not, a sensitivity analysis of key parameters, using one variable at time, was conducted. This helped to reduce the dimensionality of the history match inverse problem, making it easier to solve.

Uncertainty analysis. Uncertainty analyses were conducted using Monte Carlo sampler combined with Latin hypercube sampling, which helped to efficiently examine the combined effects of multiple uncertain parameters on objective function. Consequently, uncertainty studies capture not only individual parameter effects but also the interactions arising from joint probability distributions. The number of uncertain parameters were reduced from 16 to 10 for D field and from 11 to 8 A field, keeping the most influential

variables. This helped focusing efforts on those variables with significant impacts on target responses (Objective function).

Optimization

The Evolution Strategy (ES) optimizer was employed in the optimization process to improve the history match of the base case by minimizing the mismatch values. The ES optimizer is a derivative-free, global optimization algorithm inspired by the process of natural selection and genetics. It's particularly effective in high-dimensional spaces with non-linear relationships between parameters. The ES optimizes the parameter set by iteratively applying mutations to the current solution, thus exploring new regions of the search space. The history matching results were improved but failed to converge to global optimum, therefore machine learning was used to further minimize the objective function, [Figure 5](#).

Machine Learning – AHM

The primary ML model used in this study is a supervised learning method that uses a gradient boosting framework (Extreme Gradient Boosting regression) to predict continuous numerical values. It's a type of ensemble learning that combines multiple decision trees to create a highly accurate predictive model. hyperparameters (number of trees, learning rate, maximum depth, subsample, and column sample rate) are automatically optimized via hyperparameter optimization framework written in Python.

The outputs from multiple realizations (e.g. static and dynamic uncertainties) are consolidated as Input parameters that analyzed using parallel coordinates plots ([Figure 5](#)), Multidimensional Data Representation (MDP), enable the visualization of datasets with numerous variables by representing each variable as a parallel axis. This facilitates the identification of patterns, correlations, and outliers across multiple dimensions simultaneously.

Simulation results (multiple realizations uncertainty variables: permeability, porosity, PVT, ...etc.) together with historical production (gas, oil, and water rates) and pressure data are used as input to the machines learning models. The input data is normalized by applied Min-Max normalization to scale the data, ensuring compatibility with the ML algorithm and to prevent algorithmic bias toward large-magnitude variables.

The machine learning workflow consisted of three stages: (1) feature engineering, where relevant variables are extracted from the preprocessed data; (2) model training, where multiple ML model employed in conjunction with hyperparameter tuning using Bayesian optimization (3) prediction, where the trained ML model with dynamic simulator.

Both surrogates were trained on 80% of the dataset, reserving 20% for validation. The ML models predicted objective-function residuals—differences between measured and simulated responses—rather than raw outputs, focusing the surrogate on calibration error.

The trained proxy model can independently predict production pressure profiles for all wells, serving as a reliable substitute for the original simulation. This proxy model is integrated with an optimization system, which suggests optimal sets of uncertainty parameters based on its predictions.

The optimizer's outputs are then used to create a new reservoir simulation realization, allowing us to verify the robustness and reliability of our results.

Results Summary

ML-AHM results indicate significant improvements in model calibration, with a reduction of up to 30% and 85% in the objective for D and A fields respectively, over 100-400 ML iterations. The integrated AI-driven workflow effectively captured nonlinear relationships between reservoir parameters, leading to more accurate predictions. Overall run time to complete the ML-AHM study is five times less compared to manual history match.

Interpretability analyses provided insight into calibrated parameters. Tornado plot, and correlation plot assessment indicated that for D field: fracture permeability, followed by matrix permeability in different areas of the field were the two most influential features in reducing objective function residuals. Parameters' interaction plots further confirmed these sensitivities, guiding geoscientists to prioritize static model refinement in these areas. While for A field the most influential features are matrix permeability, transmissibility in x and y direction, pore volume multiplier and critical water saturation.

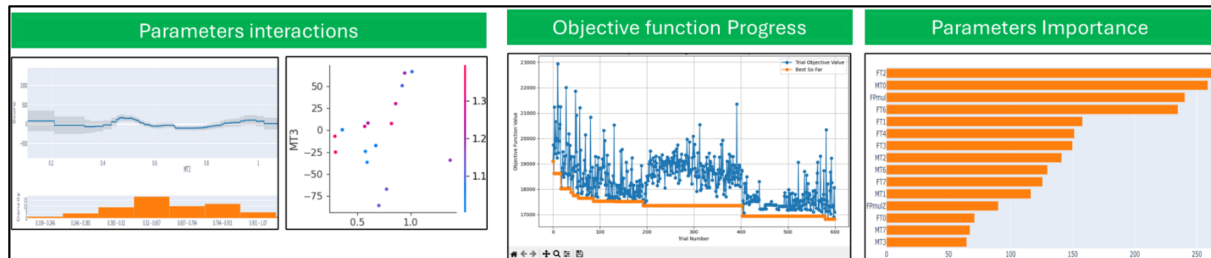


Figure 6—D field parameters importance, Convergence of objective function, and parameters interaction

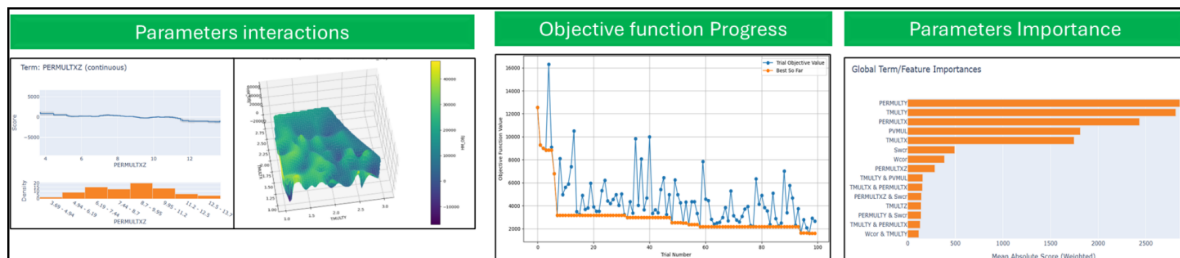


Figure 7—A field parameters importance, Convergence of objective function, and parameters interaction

The ML-AHM results showed better match as compared to manual history match for both D and A fields. In addition, field D forecast comparison revealed that manual HM case is overestimating future performance of the field as shown in figure 8.

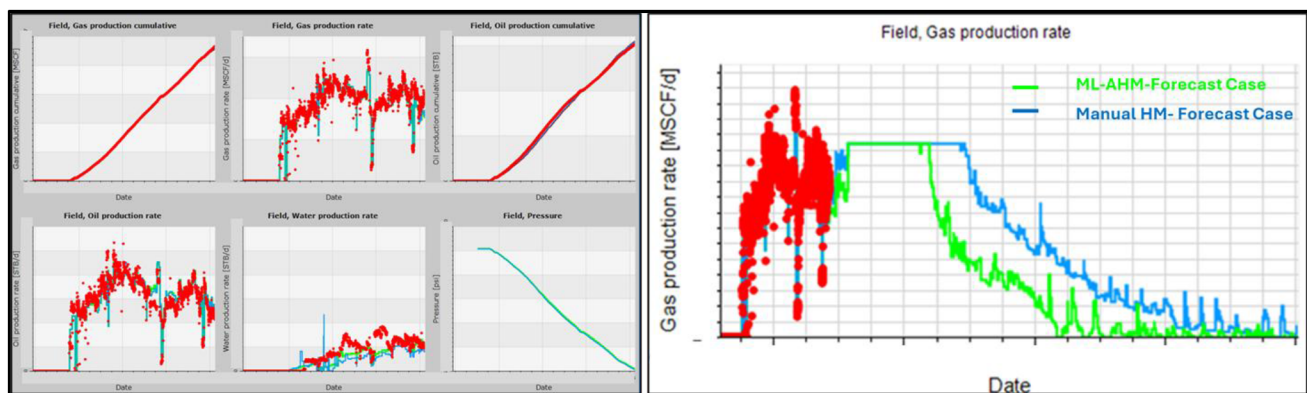


Figure 8—ML-AHM vs. Manual history match (Left figure) – Forecast Results Comparison (right figure)

These case studies revealed that data quality and model parameterization strongly influence AHM performance. For example, uncertainties in production history back allocation or uncertainties in pressure measurements or estimation can lead to multi-modal objective functions that slow convergence. It has been noted that ML-AHM can find a match "easily" but sometimes at the cost of reducing its interpretive values. Thus, rigorous quality-control is required: ML-AHM is not just curve-fitting, but a process that must preserve reservoir understanding.

Way forward Plan

By leveraging machine learning (ML)-assisted history matching outputs within a machine learning driven well placement (ML-DWP) framework, we can streamline well-placement optimization for infill wells. This integrated approach not only considers geological and operational constraints but also optimizes well's location based on a comprehensive understanding of the reservoir's uncertainty landscape. By aligning well-placement decisions with probabilistic forecasts, we mitigate the risk of suboptimal field development, thereby ensuring more informed and effective resource utilization. This synergistic approach of ML-AHM and ML-DWP can significantly enhance our ability to optimize infill wells, ultimately leading to improved reservoir performance and enhanced economic returns.

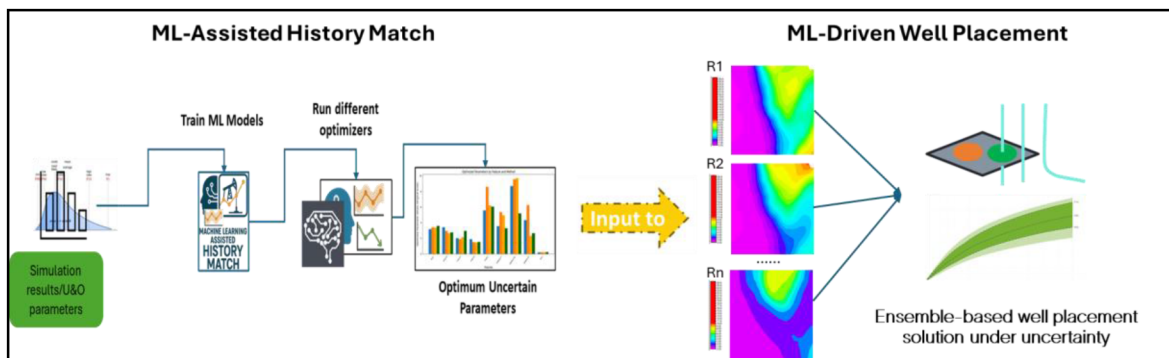


Figure 9—Integrated Approach: ML-AHM and ML-DWP

Summary, Conclusions and Future Work

Summary

This study demonstrates the potential of Machine Learning-assisted History Matching (ML-HM) in improving model calibration and predictions in North Kuwait Jurassic fields. The integrated AI-driven workflow enables quicker data assimilation and optimization of reservoir parameters, reducing uncertainty associated with traditional history matching methods.

Key Findings

- ML-HM offers several advantages over gradient-based HM, including more efficient computation and the ability to capture nonlinear relationships between reservoir parameters.
- However, ML-HM also has limitations, such as requiring additional domain-specific knowledge for effective feature engineering and model training.
- The computational cost associated with hyperparameter tuning using Bayesian optimization can be significant for large-scale applications.
- By coupling AHM with ML, the efficiency of history-matching process of field (A) and (B) was improved, achieving results up to 5 times faster compared to conventional HM process.

Conclusions

Given the intrinsic complexity of the North Kuwait Jurassic reservoirs with tight, fracture-controlled Marrat carbonates under variable subsurface uncertainties, ML-assisted history matching emerges as a transformative tool. It not only bridges the gap between vast parameter uncertainties and field-scale performance but also supports adaptive decision-making as new production data arrives. Ultimately, integrating ML into the history-matching workflow offers the clearest path to reducing reservoir-property uncertainty and unlocking the full potential of these challenging Jurassic assets.

Future Work

- Enhance computational efficiency and scalability while ensuring optimal performance of the ML-HM workflow.
- Investigate real-time data assimilation and reinforcement-learning enhancements to improve model accuracy and robustness in dynamic systems.
- Integrate machine learning-assisted history matching outputs within a machine learning-driven well placement framework to streamline well-placement optimization for infill wells.
- Scalability to complex fields: Expand the ML-AHM to more complex and bigger size fields: North Kuwait Jurassic fields, characterized by tight, fracture-controlled Middle Marrat carbonates and variable subsurface uncertainties, pose significant challenges in reservoir management. However, this study demonstrated that the integration of Machine Learning-assisted History Matching (ML-HM) can serve as a transformative tool in bridging the gap between vast parameter uncertainties and field-scale performance.
- Scalability to complex fields: Expand the ML-AHM approach to more complex and larger-sized fields characterized by tight, fracture-controlled carbonates and variable subsurface uncertainties. These reservoirs present a challenge in terms of parameter estimation and history matching which poses significant challenges in reservoir management. By adapting the ML-AHM framework to address these challenges, it can serve as a transformative tool for bridging the gap between vast parameter uncertainties and field-scale performance, ultimately facilitating more informed reservoir management decisions.

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